13

Structured cabling

Objectives

This chapter deals with structured cabling. This chapter will familiarize you with:

- Concept, objectives, and, advantages of structured cabling
- TIA/EIA standards relevant for structured cabling
- Components of structured cabling
- Concept of horizontal cabling
- Cables, outlets/connectors, patch cables etc
- Need for proper documentation and identification systems
- Likely role of fiber optic cables in coming years
- Possibility of overcoming of 100 m limits by use of collapsed and/or centralized cabling concepts
- Next generation fiber-optic cabling products

13.1 Introduction

A computer network is said to be as good as its cabling. Cabling is to a computer network what veins and arteries are to the human body.

Small networks for a few stations can be cabled easily with use of a high quality hub and a few patch cables. The majority of networks of today, however, support a large number of stations spread over several floors of a building or even a few buildings. Cabling for such networks is a different matter all together, far more complex than that required for a small network. A systematic and planned approach is called for in such cases. Structured cabling is the name given to a planned and systematic way of execution of such cabling projects.

Structured cabling, also called structured wiring system (SWS), refers to all of the cabling and related hardware selected and installed in a logical and hierarchical way.

A well-planned and well-executed structured cabling should be able to accommodate constant growth while maintaining order. Computer networking, indeed all of Information Technology, is growing at exponential rates. Newer technologies, faster speeds, more and diverse applications are becoming order of the year, if not order of the day.

It is therefore essential that a growth plan that scales well in terms of higher speeds as well as more and more connections be considered. Accommodating new technology, adding users, and moving connections/stations around is referred to as 'moves, adds, and changes' (MAC).

A good cabling system will make MAC cycles easy because it:

- Is designed to be relatively independent of the computers or telephone network that uses it; so that either can be upgraded/updated with minimum rework on the cabling
- Is consistent, meaning there is the same cabling system for everything
- Encompasses all communication services, including voice, video, and data.
- Is vendor independent
- Will, as far as possible, look into the future in terms of technology, if not be future-proof
- Will take into account ease of maintenance, environmental issues, and security considerations
- Will comply with relevant building and municipal codes

13.2 TIA/EIA cabling standards

Standards lay down compliance norms and bring about uniformity in implementations. Several vendor-independent structured cabling standards have been developed.

The Telecommunications Industry Association (TIA), Electronic Industries Association (EIA), American National Standards Institute (ANSI) and the ISO are the professional bodies involved in laying down these standards. Both the TIA and EIA are members of ANSI, which is a coordinating body for voluntary standards within United States of America. ANSI, in turn, is a member of the ISO, the international standards body.

The important standards relevant here are:

- **ANSI/TIA/EIA-568-A:** This standard lays down specifications for a Structured Cabling System. It includes specifications for horizontal cables, backbone cables, cable spaces, interconnection equipment, etc
- **ANSI/TIA/EIA-569-A:** This lays down specifications for the design and construction practices to be used for supporting Structured Cabling Systems. These include specifications for Telecommunication Closets, Equipment Rooms, and Cable Pathways etc
- **ANSI/TIA/EIA-570:** This standard specifies norms for premises wiring systems for homes or small offices
- **ANSI/TIA/EIA-606:** This standard lays down norms for cabling systems
- **ANSI/TIA/EIA–607:** This standard lays down norms for grounding practices needed to support the equipment used in cabling systems
- **ISO/IEC 11801:** This is an international standard on 'Generic Cabling for Customer Premises. Topics covered are same as those covered by the TIA/EIA-568 standard. It also includes category-rating system for cables. It lists four classes of performance for a link from class A to class D. The classes C and D are similar to Category 3 and Category 5 links of the TIA/EIA 568A standard
- **CENELEC EN 50173:** This is a European cabling standard while the British equivalent is BS EN 50173

13.3 Components of structured cabling

The TIA/EIA 568 standard lists six basic elements of a structured cabling system. They are:

Building entrance facilities: Equipment (such as cables, surge protection equipment, connecting hardware) used to connect a campus data network or public telephone network to the cabling inside the building.

Equipment room: Equipment rooms are used for major cable terminations and for any grounding equipment needed to make a connection to the campus data network and public telephone network

Backbone cabling: Building backbone cabling based on a star topology is used to provide connections between telecommunication closets, equipment rooms, and the entrance facilities.

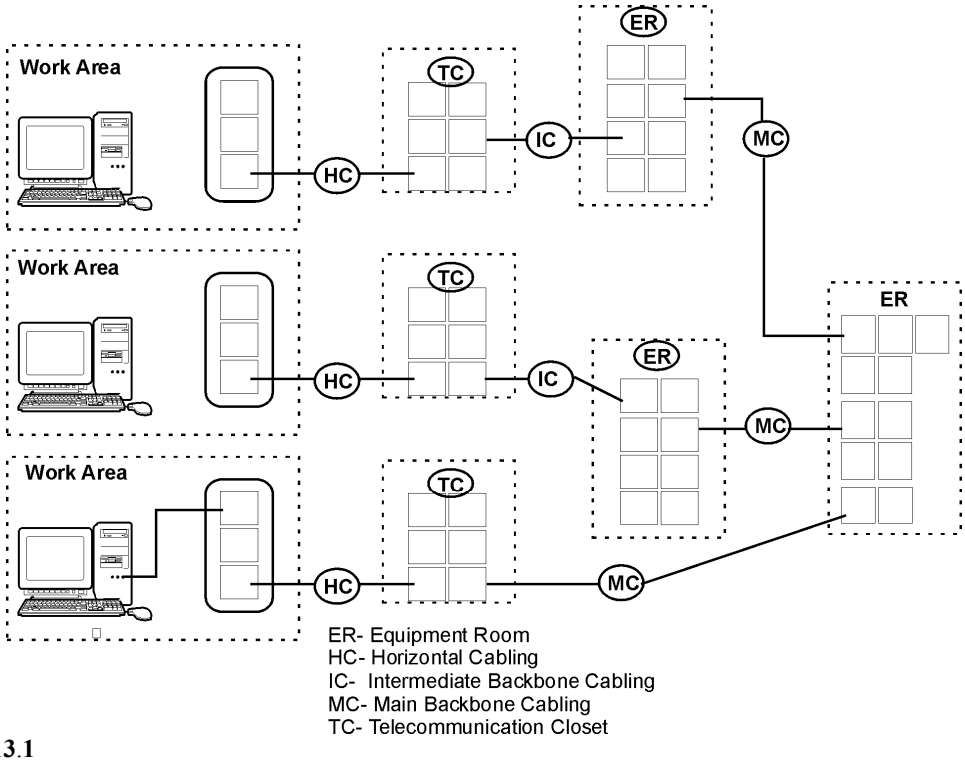


Figure 13.1
Typical elements of a structured cabling system

Telecommunication closet: A telecommunication closet, also called a wiring closet is primarily used to provide a location for the termination of the horizontal cable on a building floor. This closet houses the mechanical cable terminations and cross-connects, if any, between the horizontal and the backbone cabling system. It may also house hubs and switches.

Work area: This is an office space or any other area where computers are placed. The work area equipment includes patch cables used to connect computers, telephones, or other devices, to outlets on the wall.

Horizontal cabling: This includes cables extending from telecommunication closets to communication outlets located in work area. It also includes any patch cables needed for cross-connects between hub equipment in the closet and the horizontal cabling.

13.4 Star topology for structured cabling

The TIA/EIA 568 standard specifies a star topology for the structured cabling backbone system. It also specifies that there be no more than two levels of hierarchy within a building. This means that a cable should not go through more than one intermediate cross-connect device between the main cross connect (MC) located in an equipment room and the horizontal cross connect (HC) located in a wiring closet.

The star topology has been chosen because of its obvious advantages:

- It is easier to manage 'moves-adds-changes'
- It is faster to troubleshoot
- Independent point-to-point links prevent cable problems on any link from affecting other links
- Network speed can be increased by upgrading hubs without a need for recabling the whole building

13.5 Horizontal cabling

13.5.1 Cables used in horizontal cabling

Both twisted-pair cables and fiber-optic cables can be used in structured cabling systems.

The TIA/EIA stipulates holds that twisted-pair cables of a category better than Category 2 be used. Category 5 or better is recommended for new horizontal cable installations if twisted-pair cabling is the choice.

Specifically, TIA/EIA 568 gives the following options for use in horizontal links:

- Four-pair, 100 ohm impedance UTP cable of Category 5 or better using 24 AWG solid conductors is recommended. The connector recommended is the eight position RJ-45 modular jack
- Two-pair 150 ohm shielded twisted pair (STP) using an IEEE802.5 four-position shielded token ring connector is recommended
- Two-fiber, 62.5/125 multimode fiber optical cables are recommended. The recommended connector is the SCFOC/2.5 duplex connector 9, also known as the SC connector
- Coaxial cables are still recognized in TIA/EIA standards but are not recommended for new installations. Coaxial cabling is in fact being phased out from future versions of the standards

13.5.2 Telecommunication outlet/connector

The TIA/EIA specifies a minimum of two work area outlets (WAOs) for each work area with each area being connected directly to a telecommunication closet. One outlet should connect to one four-pair UTP cable. The other outlet may connect to another four-pair UTP cable, an STP cable, or to a fiber-optic cable as needed. Any active or passive adapters needed at the work area should be external to the outlet.

13.5.3 Cross-connect patch cables

Patch cables should not exceed 6 meters in length. A total allowance of 10 meters is provided for all patch cables in the entire length from the closet to the computer. This, combined with the maximum of 90 meters of horizontal link cable distance, makes a total

of 100 meters for the maximum horizontal channel distance from the network equipment in the closet to the computer.

13.5.4 Horizontal channel and basic link as per TIA/EIA telecommunication system bulletin 67 (TSB- 67)

TSB-67 specifies requirements for testing and certifying installed UTP horizontal cabling. It defines a basic link and a channel for testing purposes as shown in figure 13.2. The basic length consists of the fixed cable that travels between the wall plate in the work area and the wire termination point in the wiring closet. This basic link is limited to a length of 90 meters. This link is to be tested and certified according to guidelines in TSB-67 before hooking up any network equipment or telephone equipment.

Only Category 5e cables and components should be used for horizontal cabling. Components of lower category will not give the best results and may not accommodate high-speed data transfer.

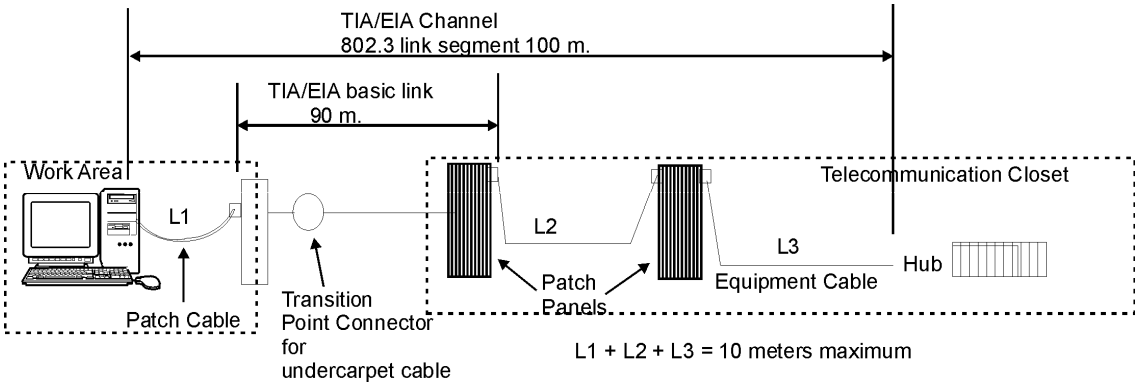


Figure13.2
Basic link and link segment

13.5.5 Documentation and identification

Even a small network cannot be organized and managed without proper documentation. A comprehensive listing of all cable installations with floor plans, locations, and, identification numbers is necessary. Time spent in preparing this will save time and trouble in 'moves, adds and changes' cycles, as well as in times of trouble shooting. Software packages for cable management should be used if the network is large and complex.

Cable identification is the basis for any cabling documentation. An identification convention suitable for the network can be set up easily. It is critical to be consistent at the time of installation as well as at times of 'moves, adds, changes' cycles.

Labels specifically designed to stick to cables should be used.

13.6 Fiber-optics in structured cabling

Future computer networks will certainly be faster, support a greater number of applications, and provide service to a number of geographically diverse users.

Fiber-optic cables are most likely to play a major role in this growth because of their unique advantages.

This sub-section is based on extracts from an article written by Paul Kopera, Anixter Director of Fiber Optics. The original article was written in 1996-97 and then revised in 2001.

'With network requirements changing constantly, it is important to employ a cabling system that can keep up with the demand.

Cabling systems, the backbone of any data communications system, must become utilities. That is, they must adapt to network requirements on demand. When a network needs more speed, the media should deliver it. The days of recabling to adopt new networking technologies should be phased out. Today's Structured Cabling System should provide seamless migration to tomorrow's network services.

One media that provides utility-like service is optical fiber. Fiber optic cabling has been used in telecommunication networks for over 20 years, bringing unsurpassed reliability and expandability to that industry. Over the past decade, optical fibers have found their way into cable television networks—increasing reliability, providing expanded service, and reducing costs. In the Local Area Network (LAN), fiber cabling has been deployed as the primary media for campus and building backbones, offering high-speed connections between diverse LAN segments.

Today, with increasingly sophisticated applications like high-speed ISPs and e-commerce becoming standard, it's time to consider optical fiber as the primary media to provide data services to the desktop.'

13.6.1 Advantages of fiber-optic technology

Fiber-optic cable has the following advantages:

- It has the largest bandwidth compared to any media available. It can transmit signals over the longest distance at the lowest cost, without errors and the least amount of maintenance
- Fiber is immune to EMI and RFI
- It cannot be tapped, so it's very secure
- Fiber transmission systems are highly reliable. Network downtime is limited to catastrophic failures such as a cable cut, not soft failures such as loading problems
- Interference does not affect fiber traffic and as a result, the number of retransmissions is reduced and network efficiency is increased. There are no cross talk issues with optical fiber
- It is impervious to lightning strikes and does not conduct electricity or support ground loops
- Fiber-based network segments can be extended 20 times farther than copper segments. Since the current structured cabling standard allows 100 meter lengths of horizontal fiber cabling from the telecom closet (this length is based on assumption use of twisted-pair cable), each length can support several GHz of optical bandwidth
- Recent developments in multimode fiber optics include enhanced glass designed to accommodate even higher-speed transmissions. With capabilities well above today's 10/100 Mbps Ethernet systems, fiber enables the migration to tomorrow's 10 Gigabit Ethernet ATM and SONET networking schemes without recabling
- Optical fiber is independent of the transmission frequency on a network. There are no cross talk or attenuation mechanisms that can degrade or limit the performance of fiber as network speeds increase. Further, the bandwidth

of an optical fiber channel cannot be altered in any manner by installation practices

Once a fiber is installed, tested, and certified as good, then that channel will work at 1 Mbps, 10 Mbps, 100 Mbps, 500 Mbps, 1 Gbps or 10 Gbps. This guarantees that a fiber cable plant installed today will be capable of handling any networking technology that may come along in the next 15 to 20 years. Thus, the installed fiber-optic cable need not be changed for the next 15 years. So rather than terming it 'upgradable', it may be termed 'future-proof'.

13.6.2 The 100 meter limit

Optical fiber specifications laid down by the TIA/EIA for structured cabling are summarized in table 13.1 below:

Optical Fiber Type	Multimode		Single Mode
Dimensions	6.25/125 microns	50/125 microns	8.3/125 microns
Bandwidth			
Low Speed (850 nm)	160 MHz-km	500 MHz.km	--
High Speed (1300 nm)	500 MHz*km	500 MHZ8km	>100 Ghz
Attenuation at			
Low Speed	3.5 dB/km	--	--
High Speed	1.5 dB/km	--	--
Backbone Cable Length	2000 m.	2000 m.	3000 m.
Horizontal Cable Length	100 m.	100 m.	Not Recommended
Applications	E-net, TR, FDDI 155 Mbps ATM Baseband Video Security Systems	E-net, TR FDDI 1.2 Gbps ATM Fiber Chanel Broadband Video	Channel Extension
Connector Type (T568-SC)	Beige Shroud	Beige Shroud	Blue Shroud

Table 13.1
TIA/EIA optical fiber specifications for structured cabling

The horizontal distance limitations of 100 m. laid down by TIA/EIA 568-A is based on the performance characteristics of copper cabling. The TIA/EIA 568-B.3 Committee is currently evaluating the extended performance capabilities of optical fiber. The objective is to take advantage of the fiber's bandwidth and operating distance characteristics to create structured cabling systems that are more robust.

13.6.3 Present structured cabling topology as per TIA/EIA

TIA/EIA 568-A recommends multiple wiring closets distributed throughout a building irrespective of whether backbone and horizontal cabling is copper or fiber based.

A network can be vertical with multiple wiring closets on each floor, or horizontal with multiple satellite closets located throughout a plant. The basic cabling structure is a star-cabled plant with the highest functionality networking components residing in the main distribution center (MDC).

The MDC is interconnected via fiber backbone cable to intermediate distribution centers (IDCs) in the case of a campus backbone, or to telecommunications closets (TCs) in case of a network occupying a single building.

From the TC to the desktop, up to 100 meters of Cat 5 UTP cable or optical fiber cable can be deployed. Typically, lower level network electronics are located in a TC and provide floor-level management and segmentation of a network. A TC also provides a point of presence for structured cabling support components, namely cable interconnect or cross-connect centers, cable storage and splices to backbone cabling.

13.6.4 Collapsed cabling design alternative

The fiber's superior performance can be used to revise the 100 m limit so that a horizontal distribution system can be redesigned to more efficiently use networking components, increase reliability and reduce maintenance and cost.

One method is to collapse all horizontal networking products into one closet and run fiber cables from this central TC to each user.

Since optical fiber systems have sufficient transmission bandwidth to support most horizontal distances, it is not necessary to have multiple wiring closets throughout each floor. With this network design, management is centralized and the number of maintenance sites or troubleshooting points is reduced. Cutting the number of wiring closets saves money and space. It reduces the number of locations that must be fitted with additional power, heating, ventilating, and air-conditioning facilities in a horizontal space. Testing, troubleshooting and documentation become easier. Moves, adds and changes are facilitated through network management software rather than patch cord manipulation. With this architecture, newly developed open-office cabling schemes (TIA/EIA TSB 75) can also be easily integrated into a network.

13.6.5 Centralized cabling design alternative

While collapsed cabling is one alternative, centralized cabling is a second alternative.
In a centralized cabling system, all network electronics reside in either the MDC or IDC. The idea is to connect the user directly from the desktop or workgroup to the centralized network electronics.

There are no active components at floor level. Connections are made between horizontal and riser cables through splice points or interconnect centers located in a TC. For short runs, a technique called fiber home run is used. It connects a workstation directly to the MDC. Low count (2 or 4 fibers) horizontal cable can be run to each workstation or office. In addition, multi-fiber cables (12 or more fibers) can support multiple users, providing backbone connections to workgroups in a modular office environment.

A centralized cabling network design provides the same benefits as a collapsed network – condensed electronics and more efficient use of chassis and rack spaces. By providing one central location for all network electronics, maintenance is simplified, troubleshooting time reduced and security enhanced. Moves, adds and changes are again addressed by software.

Centralized cabling is described by the Technical Service Bulletin, TIA/EIA TSB 72, which recommends a maximum distance of 300 meters to allow Gigabit applications to be supported.

13.6.6 Fiber zone cabling – mix of collapsed and centralized cabling approaches

This design concept is an interesting mix between a collapsed backbone and a centralized cabling scheme. Fiber zone cabling is a very effective way to bring fiber to a work area.

It utilizes low-cost, copper-based electronics for Ethernet data communications, while providing a clear migration path to higher-speed technologies.

Like centralized cabling, a fiber zone cabling scheme has one central MDC. Multifiber cables are deployed from the MDC through a TC to the user group. A typical cable might contain 12 or 24 fibers.

At the workgroup, the fiber cable is terminated in a multi-user telecommunications outlet (MUTO) and two of the fibers are connected to a workgroup hub. This local hub, supporting six to twelve users, has a fiber backbone connection and UTP user ports. Connections are made between the hub and workstation with simple UTP patch cords. The station network interface card (NIC) is also UTP-based. The remaining optical fibers are unused or left 'dark' in the MUTO for future needs.

Dark fibers provide a simple mechanism for adding user channels to the workgroup or for upgrading the workgroup to more advanced high-speed network architectures like ATM, SONET, or Gigabit Ethernet. Upgrades are accomplished by removing the hub and installing fiber jumper cables from the multi-user outlets to the workstation.

Network electronics also need to be upgraded. This process converts the network segment to a fiber home run or centralized cabling scheme. It is a very flexible and cost-effective way to deploy fiber today while providing a future migration strategy for a network. Further, an investment made in UTP-based Ethernet connectivity products is not wasted; it is, in effect, extended.

Two new cabling products have entered the marketplace, offering zone-cabling enclosures. One style mounts above a suspended ceiling, holding fiber and copper UTP cross connects, between hubs, switches, and workstations. The other style, a much larger unit, replaces a 2'×4' ceiling tile and has enough room to house a hub or other active electronics, as well as cross connects.

13.6.7 New next generation products

Over the past year, several new products have been developed that will aid in the deployment of optical fiber-to-the-desk.

To date, the standards committees are evaluating new, higher performance optical components that offer increased performance, ease of installation and lower costs. Among some of these exciting developments are small-form-factor connectors (SFFC), vertical cavity surface-emitting lasers (VCSEL) and next-generation fiber.

Advancements in fiber connectors are continuing to make fiber as viable an answer as copper.

Traditionally, fiber systems required twice as many connectors as copper cabling – crowding telecommunication closets with additional patch panels and electronics. Recently, manufacturers have introduced small-form-factor connectors that provide twice as much density as previous fiber connectors. These mini-fiber connectors hold the send and receive fibers in one housing. This reduces the space required for a fiber connection. More importantly, it decreases the footprint required on the hubs and switches for fiber transceivers. The net result is a cost reduction nearly four times to that of a conventional fiber system.

Complimenting the SFFC components are new vertical cavity surface-emitting lasers. This fiber optic transmission source combines the power and bandwidth of a laser at the

lower cost of a light-emitting diode (LED). VCSELs, when integrated into SFFC transceivers, allow for the development of higher-speed, higher-bandwidth optical systems, further extending the reach and capability of the FTTD cable system.

Next-generation fiber is 50/125 micron with a laser bandwidth greater than 2000 MHz/km at 850 nm. This fiber allows serial transmission at 10 Gigabits up to 300 meters. This next-generation fiber coupled with a 10 Gigabit, 850 nm VCSEL allows the lowest cost 10 Gigabit solutions.

Recent developments in fiber optics include:

- Enhanced glass design to accommodate high-speed transmission
- Smaller-size connectors that save space and lower cost
- Vertical cavity surface emitting laser technology for high-speed transmission over longer distances at low cost
- A vast array of new support hardware designed for Fiber Zone Cabling
- Fiber-to-the-desk is a cost-effective design that utilizes fiber in today's low-speed network while providing a simple migration strategy for tomorrow's high-speed connections. Fiber-to-the-desk combines the best attributes of a copper-based network (low-cost electronics) with the best of fiber (superior physical characteristics and upgradability) to provide unequalled network service and reliability